CODEFLARE / HACKATHON PROJECT DOSSIER

MedNexus AI

A local-first evidence review workspace, connected to a focused MRI research pipeline

ONE PROJECT | TWO LAYERS | CLEAR LIMITS

Presentation sourcebook

Product story, architecture, data flow, MRI ensemble, Grad-CAM and XAI views, reported research results, safeguards, runtime notes, judge Q&A, and a slide-ready storyboard.

Prepared 04 November 2026 | Repository and read-only service review | No fresh MRI inference performed

START HERE

The project in one minute

MedNexus is the workflow layer. NeuroVision is the task-specific MRI layer. Together, they demonstrate how different evidence can be routed to focused local services while keeping source, model output, and limitations visible.

The opportunity

People bring different kinds of evidence: MRI images, reports, and questions. A single generic assistant is not a reliable substitute for a dedicated image pipeline or a careful text review.

The product

A local-first workspace that accepts a question and one source, routes it by modality, and presents an attributed research output beside the original material.

The MRI specialist

A PyTorch pipeline for single 2D TIFF slices: a three-model classifier ensemble, a conditional three-model segmentation ensemble, a bounding box, and exploratory XAI maps.

The integration

A FastAPI app on port 8000 proxies TIFF images to the NeuroVision Flask service on port 8080. Text and ordinary images route to configured Ollama models.

The boundary

Research/demo only. Not clinically validated. The MRI result is slice-level, the model score is uncalibrated, and the full study remains for clinician interpretation.

The honest pitch

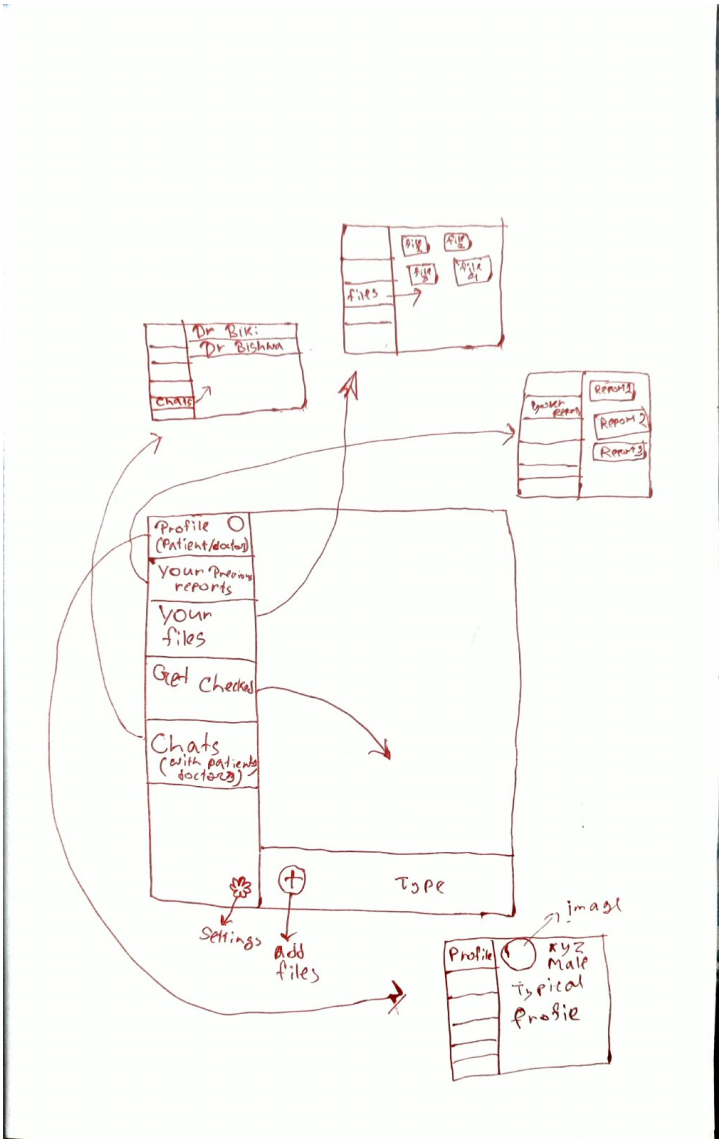
MedNexus demonstrates a modality-aware evidence handoff: route a source to a focused local model, show what came back, and keep its scope and uncertainty in view.

At the time of this review, the local MRI health endpoint reported CUDA on an NVIDIA GeForce RTX 3050 6GB Laptop GPU. The local app reported Ollama and both configured model names as available. This is a runtime snapshot, not a quality or clinical validation result.

THE IDEA

Start from a real workflow problem

The original concept sketch organizes the experience around a left-side navigation rail, a large conversation area, source uploads, and quick paths to profile, files, reports, and chats.



Original hand-drawn concept supplied for this project. The implementation turns the sketch into a persistent workspace with dedicated Chats, Profile, Files, Previous reports, Service checks, and Research safeguards views.

Problem

Evidence arrives in different forms and often gets separated from its source, model, and limitations.

Product idea

Use a shared workspace, but keep each specialist responsible for the input it can actually handle.

Design principle

Reduce the effort of getting oriented: show the source, show the route, explain the output, and make uncertainty easy to find.

Presentation framing

Pitch a careful evidence workflow and integration layer, not an autonomous diagnostic system.

CURRENT PRODUCT

What the hackathon build does today

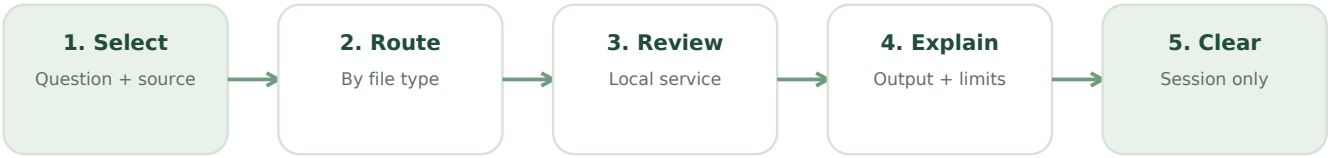
The current implementation is a working local prototype, with a separate MRI service and a browser-based MedNexus workspace.

Capability	Current implementation	Important boundary
Question + source	One question or optional context; one file per review; 12 MB upload limit.	A single upload/review flow, not longitudinal care.
TIFF MRI	Routes .tif/.tiff to NeuroVision /api/predict. Shows original slice, score, mask overlay, box data, and plain-language notes.	One 2D slice, not a DICOM series or patient-level read.
Automatic XAI	After MRI output, queues Grad-CAM, segmentor agreement, and MC-dropout variance. Click each image for its explanation popover.	Extra GPU inference; maps are exploratory model behavior views.
Medical images	PNG/JPG/WEBP/BMP routes to configured MedGemma via Ollama.	General image description is not validated for a defined clinical task.
Reports and text	PDF selectable text, TXT, MD, CSV and typed context route to configured Qwen via Ollama.	Scanned PDF OCR is not enabled; text output is preliminary.
Profile	Current-session name label, age, condition checkboxes, medications and notes. Name stays in the browser; selected health context is opt-in.	MRI model ignores the profile context; no profile database.
Service checks	Shows local MRI, Qwen, and MedGemma connectivity/model status.	Connectivity does not show model quality.
Research safeguards	Visible research notice, scope notes and readiness gates.	No regulatory clearance, clinical validation, or compliance claim.

EXPERIENCE

A review from upload to explanation

This is the judge-facing path through the current interface. Keep the original source visible and narrate each transition in plain language.



No database | UUID temporary MRI uploads | local endpoints by default

1. Ask

Write a question such as “Help me understand this MRI.” The question stays visible in the chat.

2. Attach

Choose or drop one source. The interface displays its route before submission.

3. Route

TIFF goes to NeuroVision; ordinary images to MedGemma; documents and text to Qwen.

4. Read

The MRI result explains the class label, model score, colored region, and single-slice scope.

5. Explore

XAI views load automatically. Selecting a view updates the large viewer and opens a compact descriptor.

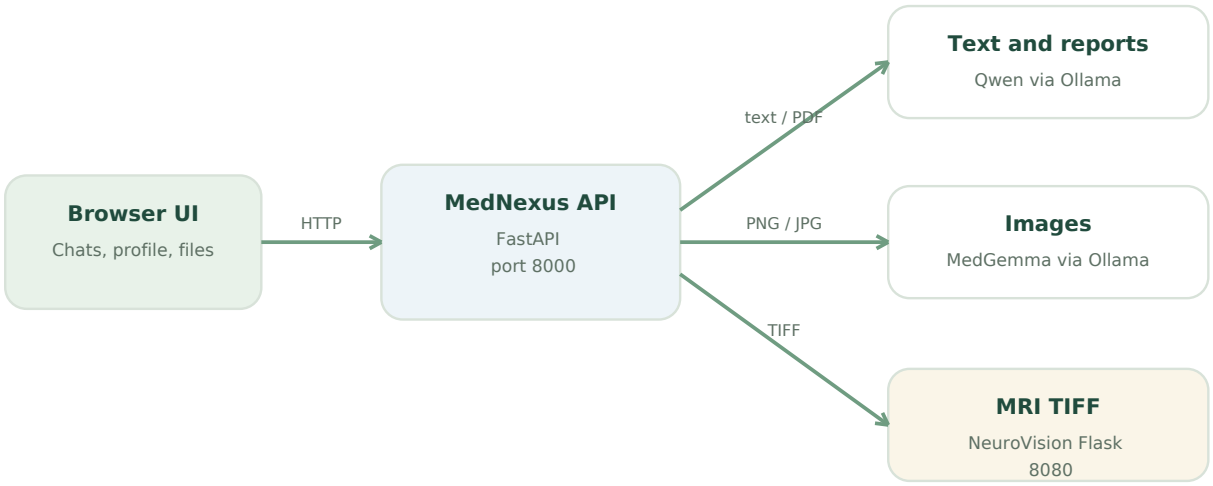
6. Continue safely

Keep the radiologist report and full MRI as the reference; use the UI to prepare questions, not treatment decisions.

SYSTEM DESIGN

Two services, one review workspace

The browser talks to MedNexus. MedNexus routes the source to a task-appropriate local service. The MRI contract remains separate from the frontend explanation copy.



Default endpoints are local; configuration can change service destinations.

Component	Role	Default address / contract
Browser UI	Vanilla HTML, CSS and JavaScript. Session state and visual review flow.	http://127.0.0.1:8000/
MedNexus API	FastAPI static shell, file routing, PDF extraction, service health, and MRI proxies.	Port 8000; /api/analyze, /api/health, /api/explain, /api/demo.
NeuroVision	Flask + PyTorch classifier/segmentor ensembles and MRI XAI.	Port 8080; /health, /api/predict, /api/explain.
Ollama	Local text and vision model API. Models unload after each request.	Default port 11434; Qwen and MedGemma model names configurable.

EVIDENCE ROUTING

A specialist for each input type

Routing is deterministic by file extension in the current app. NeuroVision is not used as a general image model, and MRI does not silently fall back to MedGemma.

Input	Route	What comes back	Current caveat
TIFF / TIF	NeuroVision MRI	Class label, raw model score, mask availability, box, overlay, scope.	One image per request; score uncalibrated.
PNG / JPG / WEBP / BMP	MedGemma through Ollama	A model-generated preliminary description.	No modality/task-specific validation established.
PDF	PyMuPDF extraction, then Qwen	Text summary and missing-context discussion.	Selectable text only; no OCR.
TXT / MD / CSV	Qwen through Ollama	Organized evidence summary from supplied content.	Output can still be wrong; clinician review required.
No file / question only	Qwen through Ollama	A response based on entered question and context.	No source means no independent evidence to review.

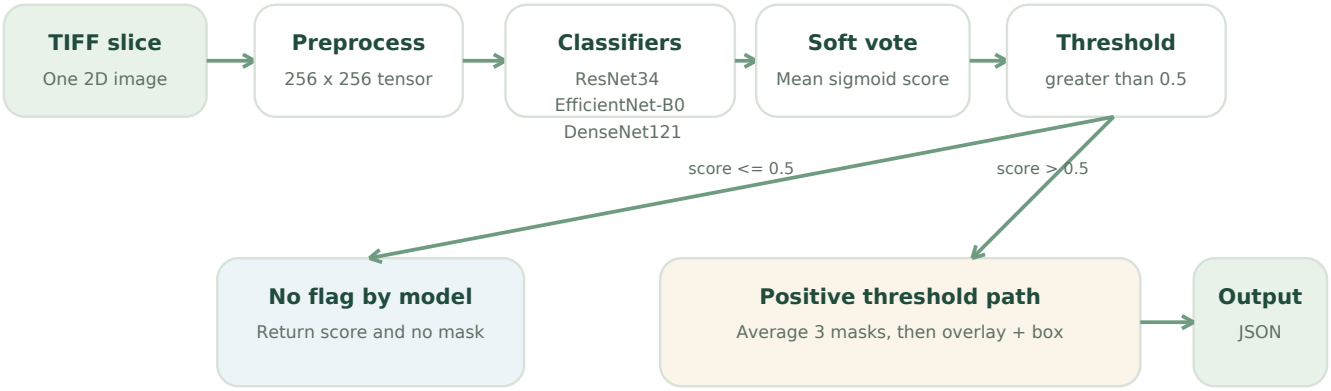
Routing is part of the story

The differentiator is not that every model can answer every question. The system keeps the selected specialist and the provenance of each output visible.

NEUROVISION

From one TIFF image to a research output

The live pipeline preprocesses one image to 256 x 256, averages the classifier ensemble, and invokes segmentation only when the classifier score exceeds the 0.5 threshold.



If model score is at or below 0.5

The service returns the class output and does not run the segmentation mask branch for this slice.

If model score is above 0.5

The three segmentors produce masks; their probability outputs are averaged and thresholded to form an overlay.

Box geometry

The implementation computes an OpenCV bounding rectangle as x, y, width, height in the transformed mask frame.

User-facing meaning

A box or colored overlay indicates model localization in the supplied image. It is not an anatomical measurement or confirmed lesion boundary.

Data boundary

NeuroVision handles an individual TIFF slice. The current path does not parse DICOM series, cross-slice relationships, or the full radiology report.

MODEL DESIGN

Two ensembles, two separate tasks

Classification asks whether the slice matches the positive class. Segmentation estimates a pixel mask. Keeping these stages distinct makes the flow easier to explain and audit.

Classifier 1

ResNet34
Residual feature backbone. Part of the equal-weight soft-voting classifier ensemble.

Classifier 2

EfficientNet-B0
Compact, compound-scaled feature model in the same classifier ensemble.

Classifier 3

DenseNet121
Dense feature connections; completes the three-model classifier vote.

Segmentor 1

U-Net + ResNet34 encoder
Predicts a mask on positive-class slices.

Segmentor 2

U-Net + EfficientNet-B0 encoder
Adds a second architecture/encoder mask estimate.

Segmentor 3

U-Net++ + DenseNet121 encoder
Third mask estimate; the ensemble averages probabilities.

Implementation note

The inference loader loads available checkpoints onto the configured device when the Flask app starts. A model that has no checkpoint is skipped with a warning; a partial ensemble can therefore differ from the intended six-model configuration.

TRAINING PIPELINE

How the research model was trained

The repository contains per-model training scripts, checkpoints, training histories, evaluation code, and a separate interactive Flask service.

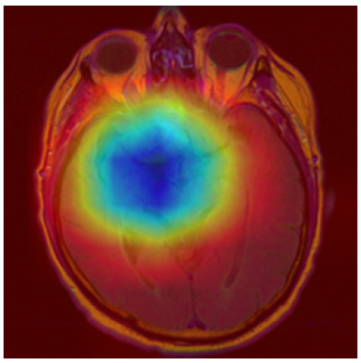
Step	Current code design	Why it matters
Data	Kaggle LGG MRI: project docs report 110 patients and 3,929 2D slices.	Small cohort; patient identity is the split unit.
Split	Deterministic patient-level split with seed 42; default 80% train, 10% validation, 10% test.	Avoids the obvious leakage risk of placing slices from one patient across splits.
Input	256 x 256 RGB image tensor; masks converted to binary.	The service expects a prepared TIFF image, not raw DICOM metadata.
Augmentation	Horizontal/vertical flip, rotation, elastic deformation, brightness/contrast.	Improves variety during training; does not replace external validation.
Classification	BCE with positive weighting; backbone frozen for epochs 1-5, then unfrozen; early stopping.	Binary class prediction from the presence of a non-empty mask.
Segmentation	Trains on positive-mask examples with Dice + BCE loss; early stopping.	Optimizes overlap of the predicted mask with the reference mask.
Precision	Mixed precision setting enabled in shared config.	Training/runtime detail; not evidence of clinical performance.

Checkpoints currently present in the repository total about 1.1 GB on disk. Disk size is not a VRAM estimate. The active service reports a 6 GB RTX 3050 laptop GPU.

VISUAL EVIDENCE

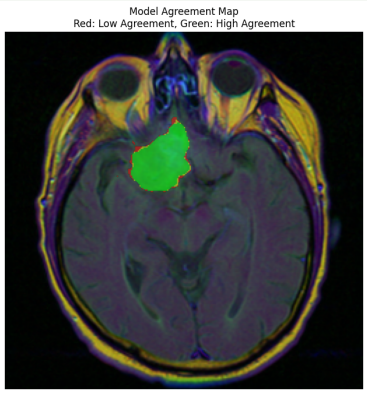
Three XAI views, explained carefully

These existing images are saved project outputs. They were not generated during this PDF build. In the app, they load automatically after an MRI result and each image opens a plain-language descriptor.



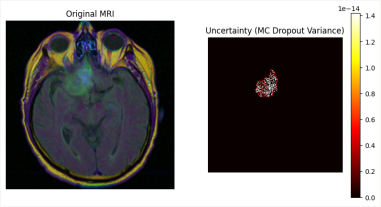
Grad-CAM

Warmer regions were associated with the first classifier model's tumour-category output in that pass. This is not a tumour outline or a cause.



Segmentation agreement

Green indicates three-model overlap; yellow two; red one. Agreement is model consistency, not correctness.



MC-dropout variation

Shows variation over repeated passes of one segmentor. Not a calibrated probability or validated uncertainty measure.

A visual is not a clinical explanation

Grad-CAM is generated from one classifier model; agreement compares segmentation model masks; the variance view comes from repeated passes of one segmentor. None is a diagnosis, validated explanation, or clinical certainty estimate.

Sample filename family: sample_tumor_*.png. Keep sample images clearly identified as examples, not new live inference or patient-specific assessment.

USER-FACING EXPLANATION

Translate outputs without pretending to read the full scan

The new result panel leads with what a person can take away from the model output, explains the colored area, and avoids treating raw coordinates as clinically meaningful.

MODEL FLAG	MODEL SCORE	SHADED REGION
“The model found a pattern in this one image that it learned to associate with low-grade glioma, so it flagged the slice for review. This is not confirmation of a tumor.”	“0.984 is a tumour-category model score, not a 98.4% personal chance.”	“The colored area is where a separate segmentation model marked pixels. Compare it with the original; it is not a confirmed boundary.”

A useful next question

“What does the radiology report say about this area, and how does it fit with the rest of my MRI?”

Why this is more useful

It translates the model’s class, score, and mask into ordinary words and identifies what to ask about.

Why it stays bounded

It never claims to identify the patient’s diagnosis. It directs interpretation back to the complete MRI and radiologist report.

XAI card: classifier focus

Warm regions are more associated with this run’s output; not a lesion outline, probability map, or causal explanation.

XAI card: agreement / variation

Agreement shows overlap between model masks. Variation shows differences over repeated passes. Neither establishes correctness.

INTERFACE MAP

The pages judges can navigate

The UI uses a shared workspace and separate destinations, with the research notice visible while reviewing.

Page	What it contains	Use in the demo
Chats	Question composer, context disclosure, one-file attachment, drag/drop, progress and result cards.	Start a review and show the working flow.
Profile	Name label stored in the browser UI; optional age, diabetes, blood pressure, neurological history, surgery, medication, notes.	Show session-only intake. Name is not sent; MRI model ignores the medical context.
Your files	Current session source list, file size, route and review status.	Show provenance and temporary working set.
Previous reports	Outputs, source names, time, route and session scope.	Show the current session, not a durable chart.
Service checks	NeuroVision, MedGemma, Qwen status and observed device.	Prove that services are connected without implying quality.
Research safeguards	Prototype status, limits, and pre-pilot evidence/governance gates.	Explain what has not yet been established.

Persistent notice

Research prototype; not for clinical use. Model outputs are preliminary and unvalidated. They do not diagnose or guide treatment. Review source material with a qualified clinician.

INTEGRATION CONTRACT

The API boundary keeps the MRI service focused

MedNexus adapts the MRI response for a human-readable workspace without changing the NeuroVision /api/predict JSON contract.

Endpoint	Owner	Purpose
GET /	MedNexus	Serve the HTML application.
GET /api/health	MedNexus	Check Ollama tags, configured models, and MRI service health/device.
POST /api/analyze	MedNexus	Accept question/context and optional file; validate size/type; route by modality.
POST /api/explain	MedNexus proxy	Relay a TIFF to NeuroVision’s additional XAI route.
GET /api/demo	MedNexus	Return clearly marked synthetic example content.
GET /health	NeuroVision	Expose loaded device and service identity.
POST /api/predict	NeuroVision	Accept TIFF multipart field `file`; return prediction, uncalibrated score, mask/box and scope.
POST /api/explain	NeuroVision	Return temporary original/Grad-CAM/agreement/variance data URLs.

Failure behavior

If NeuroVision is unavailable or rejects a TIFF, MedNexus reports a service error. It does not silently route the MRI to a generic vision model.

RUNTIME

Local services and GPU policy

The current run is configured for local services. The MRI app now requires CUDA and fails at startup if the GPU is unavailable; it will not fall back to CPU.

8000	8080	CUDA	6 GB	12 MB
MedNexus	NeuroVision	MRI device	RTX 3050	upload limit

MedNexus

FastAPI/Uvicorn. Static frontend and JSON API. Text and image requests call local Ollama with keep_alive=0.

NeuroVision

Flask app loads available classifier and segmentor checkpoints onto CUDA and serializes model work with a lock.

Memory note

Six checkpoint files total about 1.1 GB on disk. Runtime memory includes model allocations and activations and is higher.

Operating note

The XAI route adds GPU work after the base MRI response. It may take longer; a failed XAI pass should not hide the base result.

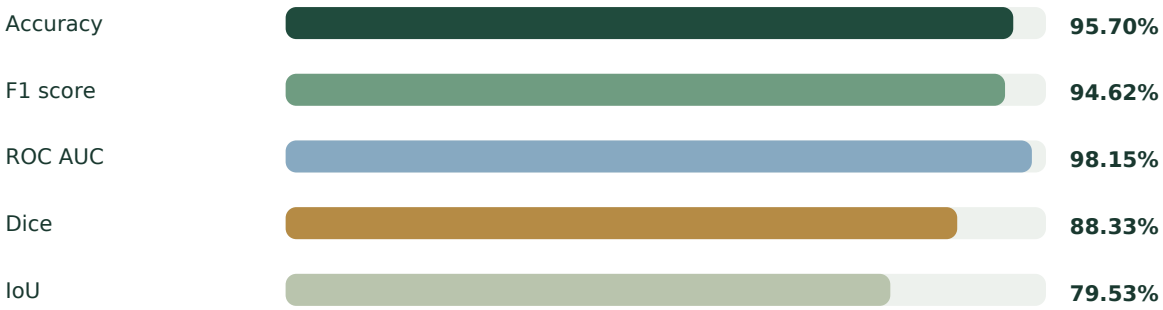
Last observed status on 04 Nov 2026

NeuroVision /health reported status ok, device cuda, NVIDIA GeForce RTX 3050 6GB Laptop GPU. MedNexus /api/health reported the MRI service and both configured Ollama model names available. Health is connectivity metadata only.

RESEARCH EVIDENCE

Project-reported performance

The original MRI project report records results on its patient-held-out test split. The figures below are transcribed from that report, not freshly recomputed for this dossier.



Separate task metrics; classification and segmentation are not one shared score.

Task	Metric	Reported value	Source definition/context
Classification	Accuracy	95.70%	Binary class on 349-slice unseen-patient test set, per report.
Classification	Precision	96.15%	Reported in the project PDF.
Classification	Recall / sensitivity	92.59%	Reported in the project PDF.
Classification	F1 score	94.62%	Reported in the project PDF.
Classification	ROC AUC	98.15%	Reported in the project PDF.
Segmentation	Average Dice	88.33%	Positive-mask evaluation; report describes average score.
Segmentation	Average IoU	79.53%	Positive-mask evaluation; report describes average score.

Use the correct label

Say “project-reported research-set results.” Do not call these clinical accuracy, a probability for a person, or proof of generalization to other hospitals/scanners.

INTERPRET THE RESULTS

What the numbers do and do not show

A good presentation makes the method and evidence legible while showing that research metrics are not the same as clinical readiness.

What is documented

The report describes patient-level splitting, 110 patients, 3,929 slices, and a 349-slice held-out test set.

What the current score is

A mean of sigmoid outputs from the available classifier models, thresholded at 0.5. It is explicitly marked uncalibrated.

What Dice / IoU mean

Overlap between predicted and reference segmentation masks under the evaluation code; computed on positive-mask batches.

What remains unknown

External-site performance, scanner and protocol robustness, calibration, subgroup performance, prospective workflow impact, failure rates and clinical utility.

Documentation discrepancy

Legacy project text describes a 4 GB target; current runtime reports an RTX 3050 6 GB. Present the observed hardware accurately.

Reproducibility note

The metric values are sourced from the report and README. This dossier did not rerun training or live inference.

A strong judge answer

“We have an internal held-out evaluation from the research dataset. We have not established external clinical validation, calibration, or suitability for patient care.”

TRUST AND GOVERNANCE

Trust signals are visible; compliance is not claimed

The product emphasizes provenance and boundaries. Those are useful design choices, not proof of legal or regulatory compliance.

Built into this prototype	Still to establish before any patient-facing pilot
Research-only notice on the workspace.	Intended use, intended users, jurisdiction, and regulatory classification review.
Scores labeled uncalibrated; no diagnosis/treatment language in system prompt.	Independent representative clinical validation, calibration and subgroup/error analysis.
MRI, text, and image routes remain separate; MRI fails rather than silently falling back.	Human factors and automation-bias study with the intended users.
UUID temporary MRI uploads and finally cleanup in the MRI endpoints; local default endpoints.	Data-flow inventory, access control, legal basis/consent, retention, deletion, incident response, backup and audit strategy.
Original scan remains available beside model views.	Defined operating procedures, escalation, accountability, monitoring and change control.

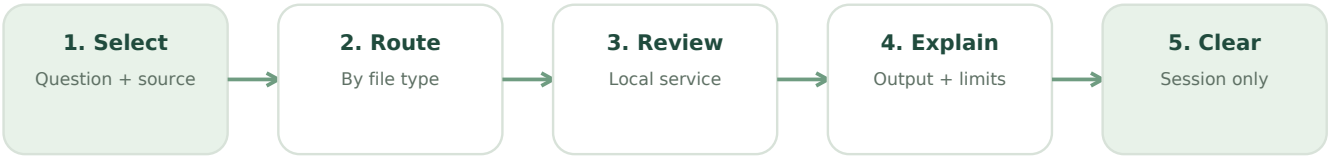
Presentation wording

Describe this as a research/hackathon prototype with safeguards and a readiness plan. Do not say “HIPAA compliant,” “FDA cleared,” “clinically trusted,” or “diagnostic” unless independently established for the actual deployment and intended use.

DATA HANDLING

Why the MVP has no database

Session-only state keeps the prototype simple and avoids creating a permanent store of health information before the product has an access, retention, and governance model.



No database | UUID temporary MRI uploads | local endpoints by default

Current

No database, login, durable patient chart, or cross-device resume. Chat, files and report objects exist in browser memory for the current page session.

Profile name

Shown in the current browser screen and omitted from the model request.

Age and conditions

Sent only after the user chooses to add context and submits a review. The MRI prediction route does not use these fields.

Temporary files

MRI uses unique UUID names and `finally` cleanup. Generic PDF extraction is performed in request memory; PDF OCR is not enabled.

When a database becomes appropriate

Add persistence only for a clear requirement such as consented user accounts, durable reports, team review, or audit history. Pair it with identity/access controls, encryption, retention/deletion, audit logging, backup and incident response. A database alone does not create compliance.

PRESENTER RUNBOOK

Before the judges arrive

Keep the demo path predictable, use approved de-identified samples, and prepare a fallback that is clearly labeled as synthetic.

Order	Action	What to say / check
1	Start NeuroVision with MRI_DEVICE=cuda.	Confirm /health says device cuda and the intended GPU. Never change to CPU.
2	Start MedNexus on port 8000.	Open the workspace and wait for health status.
3	Confirm local model services.	Qwen and MedGemma names should appear; health only proves service response.
4	Choose one sample TIFF.	Use the project sample. Avoid real patient identifiers.
5	Submit a short user question.	Point out TIFF-specific routing and source-first original image.
6	Wait for the XAI pass.	Explain Grad-CAM, mask agreement and variation in plain language; never call them a diagnosis.
7	Close on the safeguard.	Return to the full report/scan and non-clinical-use notice.

Fallback

If a live specialist is unavailable, use “See an example” and explicitly say the output is synthetic demo data. Do not pass a saved visualization off as the result of a new live inference.

Pre-demo caution: the automatic XAI stage adds GPU work after the MRI result. Leave time for it, keep Ollama concurrency modest, and keep the base MRI result visible if an XAI route fails.

PITCH

A concise 90-second walkthrough

The pitch works best when every claim can be pointed to in the running UI or the repository.

0:00-0:08 | Problem

“Health evidence arrives in different forms. A general chat model should not improvise MRI segmentation.”

0:08-0:20 | Routing

Select a TIFF; show that MedNexus routes it to NeuroVision rather than a general image model.

0:20-0:40 | Evidence

Show original slice, model output, uncalibrated score, and the separate segmentation overlay.

0:40-1:00 | XAI

Open Grad-CAM and agreement/variation cards. State what each describes and what it cannot prove.

1:00-1:12 | Boundary

Point to the research notice and full-study limitation; say the report and clinician remain the reference.

1:12-1:25 | Wider workflow

If available, submit a text report or ordinary image to demonstrate separate routing.

1:25-1:30 | Close

“The useful part is the evidence handoff: the right specialist, visible source, and clear limits in one workspace.”

One-line pitch

MedNexus routes health evidence to configured local specialists and keeps the source, model output, and limitations together for review.

JUDGE Q&A

Short, defensible answers

Keep answers specific. If something is planned rather than implemented, say that plainly.

What is novel?

A modality-aware local evidence workflow with a task-specific MRI service and a shared source/provenance UI.

Does it diagnose brain tumors?

No. It returns preliminary research-model output for one 2D slice and is not clinically validated.

What does 0.984 mean?

It is the classifier ensemble’s uncalibrated output score for its tumor-category class, not a 98.4% chance for the patient.

Why use an ensemble?

Three classifier outputs are averaged for classification; three segmentation outputs are averaged for a mask. This is a design choice, not proof the ensemble is clinically superior.

What does Grad-CAM prove?

Nothing clinical. It visualizes regions associated with one classifier’s output in a pass; it is not causal evidence or a lesion boundary.

Why no database?

The MVP has no accounts or durable history. Session-only storage minimizes what the demo persists until access, retention and audit needs are designed.

Is it private?

Default service endpoints are local, and temporary MRI uploads are cleaned up. Deployment configuration and machine access still matter; do not use real identifiers in the demo.

What is next?

Define intended use, validate externally, evaluate calibration and subgroup performance, study human factors, and design governed persistence only if needed.

NEXT STEPS

A credible path beyond the hackathon

A future clinical product requires evidence, human workflow design, governance, and robust engineering. The roadmap below is proposed work, not completed capability.

Stage	Focus	Evidence / deliverable
1. Stabilize prototype	Responsive UI, errors, service startup, GPU guard, fixture API checks, reproducible sample flow.	Versioned demo release; known-issues list; end-to-end scripted walkthrough.
2. Define intended use	Specify users, population, setting, clinical purpose, inputs, outputs, exclusions and jurisdiction.	Intended-use statement and expert classification review.
3. Validate model	Independent cohort and sites, predefined endpoints, calibration, subgroup analyses, external failure review.	Study protocol, evaluation report, error taxonomy, uncertainty limitations.
4. Study workflow	Test interface with intended clinicians; measure comprehension, automation bias, escalation and overrides.	Human-factors plan and documented workflow/accountability.
5. Govern deployment	Security model, consent/lawful basis, access, retention, deletion, audit, incidents, backups, monitoring.	Approved deployment controls and post-release monitoring plan.
6. Expand data support	DICOM series support, metadata handling, series-level context, source-report linking.	Validated ingestion and traceability specification before implementation.

Do not leapfrog evidence

A persistent database, cloud deployment, or more persuasive XAI display does not establish clinical readiness. Add each capability only with its safety and governance design.

PPT CONVERSION

Slide-ready storyboard / 1

A suggested 22-slide sequence. The listed visual is a cue for building slides; it does not imply that a clinical capability exists beyond the current prototype.

#	Slide title	Takeaway	Visual cue
1	Title	MedNexus AI: a modality-aware evidence review workspace	Product name + one-line pitch
2	The problem	Health evidence is fragmented across scans, reports and questions	Simple source-type collage
3	The product idea	A shared workspace with specialist routing by evidence type	Original sketch and current page map
4	The user journey	Ask, attach, route, inspect, explain	Five-step flow diagram
5	System architecture	Browser -> FastAPI -> local specialists	Architecture diagram
6	Routing logic	TIFF, ordinary image, PDF and text take distinct paths	Routing matrix
7	MRI dataset and task	One 2D slice; LGG dataset described in the project report	Dataset facts + scope label
8	Classifier ensemble	ResNet34 + EfficientNet-B0 + DenseNet121 soft vote	Model cards
9	Segmentation ensemble	Three U-Net-family segmentors; mean mask on positive path	Model cards + overlay
10	MRI output	Class label, uncalibrated score, mask and pixel box	Original vs overlay visual
11	XAI: classifier view	Grad-CAM association, not causal evidence	Grad-CAM visual and legend

Deck style

Use large, legible labels; one main idea per slide; show the original scan beside overlays; label research metrics as project-reported; keep “not for clinical use” visible.

PPT CONVERSION

Slide-ready storyboard / 2

A suggested 22-slide sequence. The listed visual is a cue for building slides; it does not imply that a clinical capability exists beyond the current prototype.

#	Slide title	Takeaway	Visual cue
12	XAI: mask agreement	Overlap among segmentor outputs, not correctness	Agreement map color key
13	XAI: variation	MC-dropout variation is not calibrated clinical uncertainty	Variation map + caveat
14	Plain-language UX	Explain what output says and what it does not	Result card screenshot or copy mockup
15	Reported research metrics	Project PDF numbers on held-out research slices	Metric chart with source caveat
16	Evidence boundaries	Not external clinical validation or a personal risk probability	What is known / unknown table
17	Privacy and persistence	Local defaults, temporary processing, no database in MVP	Data lifecycle diagram
18	GPU-first runtime	CUDA-only MRI path; current runtime GPU snapshot	Service status card
19	Safeguards	Visible limits and readiness gates	Safeguard page / release-gate cards
20	Demo walkthrough	One sample TIFF, auto XAI, report-level question	Live flow
21	Roadmap	Intended use, independent validation, human factors, governance	Maturity roadmap
22	Close	The evidence handoff is the product	Source + model + limitations

Deck style

Use large, legible labels; one main idea per slide; show the original scan beside overlays; label research metrics as project-reported; keep “not for clinical use” visible.

REPOSITORY MAP

Where the project lives

The repo contains the research model, its web service, the MedNexus integration, and the test suite. The MRI model and the evidence workspace are separate modules.

Path	Purpose
Brain MRI segmentation/app.py	Flask runtime, GPU-only setup, /health, /api/predict, /api/explain, legacy HTML.
Brain MRI segmentation/src/models/	Classifier and segmentor wrappers.
Brain MRI segmentation/src/inference.py	Soft-vote classification, threshold, conditional mask, bounding box, checkpoint loading.
Brain MRI segmentation/src/explainability/	Grad-CAM, segmentation agreement, MC-dropout, map generation.
Brain MRI segmentation/src/train_*.py	Independent classifier and segmentor training scripts.
Brain MRI segmentation/report.pdf	Long-form project report; source of reported research metrics and dataset description.
mednexus-ai/server/main.py	FastAPI app, file routes, prompt boundaries, health and MRI proxy routes.
mednexus-ai/web/	Single-page interface: chats, profile, files, reports, service checks and safeguards.
mednexus-ai/integration/	MRI JSON bridge and integration notes.
tests/e2e/	101 API-contract, edge-case, combination and scenario tests; fixture mode avoids model inference.
CODEX_HANDOFF.md	Project integration notes and operational constraints.

Preserve source assets

The project contains existing dataset/sample images and saved XAI visualizations. Keep sample/test assets clearly labeled and do not present a saved sample as a fresh live result.

APPENDIX

Plain-language glossary

Use the first column to translate model terms for a mixed technical/non-technical judging panel.

Term	Plain-language meaning
2D slice	One image from a larger MRI study. It is not the full scan.
Classifier	Model that assigns the image to a category.
Soft vote	Average of classifier sigmoid outputs in the current ensemble.
Score	Raw model output. It is explicitly not calibrated as a probability for a person.
Segmentation mask	Pixel map that marks an area selected by a segmentation model.
Bounding box	A rectangle around the mask in the transformed image frame; not a tumor size or severity measure.
Grad-CAM	A heatmap associated with one model’s output; not causality or a lesion outline.
Agreement map	A pixel-level view of how many segmentation models marked an area.
MC dropout	Repeated stochastic model passes used to visualize prediction variation.
Dice / IoU	Metrics comparing overlap between predicted and reference masks.
DICOM	A medical imaging format and metadata standard; full DICOM-series ingestion is not the current product path.
Local-first	Default service endpoints run on the same machine; this is not a guarantee about every possible configuration.

Words to avoid in the presentation

“AI radiologist,” “diagnosis,” “clinically validated,” “98.4% chance,” “tumor boundary confirmed,” and “compliant” are not supported by this prototype.

APPENDIX

Source register and preparation notes

This dossier was assembled from code, project documentation, saved output images, and read-only service metadata in the shared workspace.

Ref	Source	Used for
[S1]	Brain MRI segmentation/report.pdf	Dataset description; evaluation split and model metric values.
[S2]	Brain MRI segmentation/README.md	Project overview, architecture summary and headline metrics.
[S3]	Brain MRI segmentation/src/config.py; src/inference.py; src/ensemble.py	Model list, input size, ensemble behavior, threshold and bounding box.
[S4]	Brain MRI segmentation/src/explainability/compare.py; src/explainability/segmentation_xai.py	Grad-CAM target model, agreement colors, MC-dropout pass count.
[S5]	Brain MRI segmentation/app.py	GPU-only setup, endpoints, upload cleanup, CUDA health information.
[S6]	mednexus-ai/README.md; DEMO_SCRIPT.md; integration/README.md	Product overview, routing, demo story and integration boundaries.
[S7]	mednexus-ai/server/main.py	API routing, prompt boundaries, local model calls, health and file size.
[S8]	mednexus-ai/web/index.html; app.js; styles.css	Current pages, profile behavior, explanation copy, XAI interaction and safeguards.
[S9]	tests/e2e/ and TEST_INFRA.md	Fixture suite structure and test counts.
[S10]	Read-only local health response, 04 Nov 2026	Observed service reachability, model names, CUDA device and GPU name.
[S11]	Original user concept sketch	Navigation and chat composition concept.

Preparation limits: no MRI model was run for this document. The MRI/XAI visuals are existing saved samples. Reported metrics were transcribed from project documentation and not independently recomputed. The browser UI was not captured as a screenshot for this dossier; the project experience is described from source code and user-provided comments. The document is a project/presentation aid, not a clinical or regulatory assessment.

END OF DOSSIER